ST. JOSEPH'S COLLEGE OF ENGINEERING

Department of Computer Science and Engineering (Cyber Security) · OMR, Chennai

R&D IDEA PROPOSAL FORMAT

Hackathon Idea Submission

Project / Idea Title	
Team Name	
Hackathon Name	
Problem Statement ID / Theme	
Domain / Category	☐ Software ☐ Hardware ☐ Both
Faculty Mentor	
Date of Submission	

T Team Details

#	Name	Register No.	Year / Sec	Role (Lead / Dev / Design / Research)
1				
2				
3				
4				
5				
6				

1 Problem Statement

What real problem are you solving? Who faces it, how often, and what does it cost them today? State it in 3–4 sentences.

Problem · Who is affected · Current impact

2 Background & Existing Solutions

What already exists (products, research papers, open-source tools)? Name at least 3 and say why each falls short.

Existing solution → Limitation

3 Objectives

List 3–5 specific, measurable goals. Example: “Detect phishing URLs with ≥ 95% accuracy in under 1 second.”

1. 2. 3.

4 Proposed Solution

Describe your idea and how it works. Attach or sketch a block / architecture diagram.

Idea · How it works · Architecture sketch

5 Novelty / Innovation

What is new or different about your approach compared to Section 2? This is the most important section for evaluators.

What makes it unique

6 Methodology & Technology Stack

Step-by-step plan for building it. Mention languages, frameworks, datasets, hardware and APIs you will use.

Steps · Tools · Dataset · Hardware

7 Feasibility

Can your team build a working prototype within the hackathon timeline? What skills do you have, and what will you need to learn?

Technical · Time · Skills

8 Expected Outcomes & Impact

What will the final prototype demonstrate? Who benefits, and how will you measure success?

Deliverable · Beneficiaries · Success metrics

9 Timeline & Milestones

Break the work into phases with owners and target dates.

Phase	Activity	Owner	Target Date
Phase 1	Research & problem validation		
Phase 2	Design & architecture		
Phase 3	Prototype development		
Phase 4	Testing & evaluation		
Phase 5	Final demo & documentation		

10 Risks & Mitigation

What could go wrong (data unavailable, accuracy too low, integration issues)? What is your backup plan?

Risk → Mitigation

11 Resources / Budget (if any)

Hardware, cloud credits, datasets, licences or components required, with approximate cost.

Item · Quantity · Approx. cost

12 References

Papers, websites, datasets and tools cited above (IEEE style preferred).

[1] [2] [3]

✓ Submission Checklist

☐ All 12 sections filled	☐ Architecture diagram attached	☐ References cited
☐ Team roles assigned	☐ Novelty clearly stated	☐ Mentor reviewed

R Mentor Remarks

Recommendation: ☐ Approved ☐ Revise & resubmit ☐ Not approved

Team Lead	Faculty Mentor	Head of Department